

CI / CD

What is CI/CD?

CI/CD stands for Continuous Integration and Continuous Delivery (or Continuous Deployment). It is a method used to frequently deliver apps to customers by introducing automation into the stages of software development.

It is like an automated assembly line for your code. Instead of manually building, testing, and deploying your software, a CI/CD pipeline does it automatically every time a developer changes the code.

1. Continuous Integration (CI)

Whenever a developer pushes new code to the repository (like Git), the CI system automatically grabs the code, builds the project, and runs automated tests.

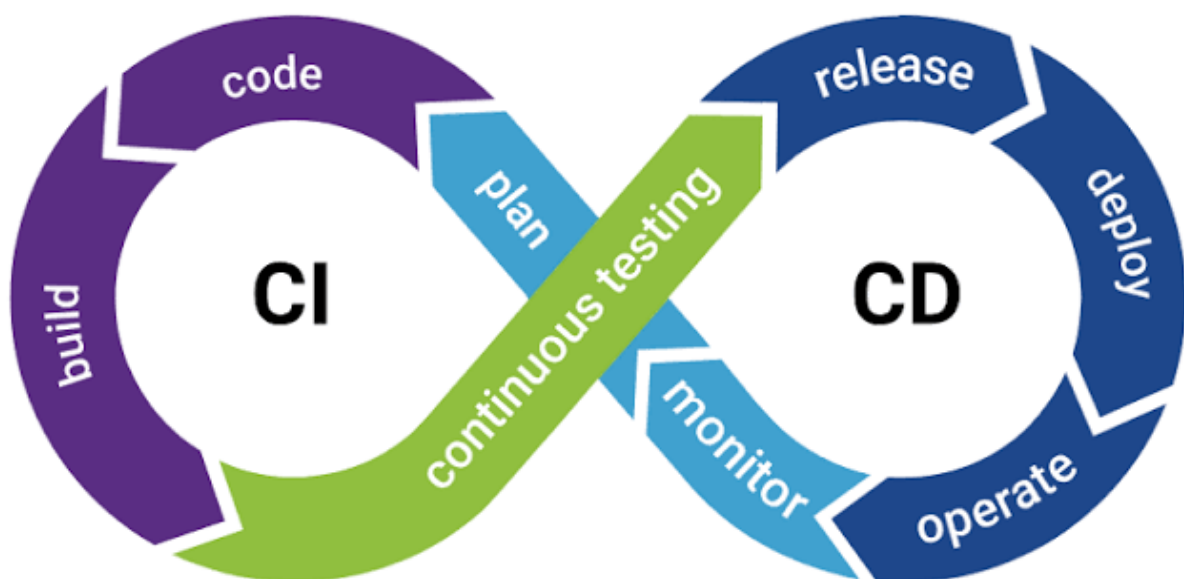
The Goal: Catch bugs early. If someone writes broken code, the build fails immediately, alerting the team before the issues spread.

2. Continuous Delivery / Deployment (CD)

Once the code passes all the CI tests, CD takes over.

Continuous Delivery: Automatically prepares the code to be deployed, but waits for a human to click "Approve" before pushing it to production.

Continuous Deployment: Takes automation a step further. Every change that passes the tests is automatically deployed straight to live users without any manual intervention.



Advantages of CI/CD Pipelines.

- **No More Manual Installs:** You don't have to manually drag and drop files or run terminal commands to deploy code; the pipeline handles it.
- **Smaller, Safer Changes:** Because deploying is so easy, teams push small, frequent updates rather than massive, risky updates once a month.
- **Faster Bug Detection:** Automated testing catches errors within minutes of code being written, making them much easier to fix.
- **Consistent Environments:** The pipeline deploys the software the exact same way every single time, eliminating the classic "it works on my machine" problem.